

JIAHE JIANG

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EDUCATION

Bachelor of Computer Science, The University of British Columbia
Globex Julmester program, Peking University
Cumulative GPA: 4.10/4.33

Expected Dec 2026
Jul 2024

PUBLICATIONS

Jiang, J., Ding, C., Zheng, W., & Zhai, X. (2026, April). Model-Driven Aided Expectation Propagation Learning for AFDM Signal Detection in High-Mobility Communications. In 2026 IEEE Wireless Communications and Networking Conference Workshops (WCNCW) (pp. 1-6). IEEE.

He, X., Yang, T., Yan, T., Li, H., Ge, Y., Ren, Z., ... & Tang, C. (2026). Efficient Layer-Wise Cross-View Calibration and Aggregation for Multispectral Object Detection. *Electronics*, 15(3), 498.

C. Duan, L. Xiao, J. Jiang, J. Zhou, P. Xiao and T. Jiang, "ZC-OFDM Aided Waveform Design for LEO Satellite Integrated Communication and Navigation," in *IEEE Wireless Communications Letters*, doi: 10.1109/LWC.2025.3620133

C. Tang et al., "Multi-View Clustering via High-Order Bipartite Graph Learning and Tensor Low-Rank Representation," in *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 11, pp. 6521–6534, Nov. 2025, doi: 10.1109/TKDE.2025.3603594

RESEARCH INTERESTS

Trustworthy machine learning, multimodal representation learning, computational mass spectrometry, generative modeling, signal processing, and computational biology

RESEARCH EXPERIENCE

Research Assistant

Jan 2026 – Present

UBC Department of Chemistry – Huan Lab

On-site

- Developed computational workflows for DIA deconvolution and IodoFinder-based LC-MS/MS analysis, integrating blank correction, feature filtering, spectral-library construction, and visualization to support single-cell metabolomics and unknown-compound discovery
- Built a React/FastAPI platform to replace fragmented analysis steps, automating data upload, job tracking, result retrieval, and interactive review of mass spectrometry outputs
- Recognized that the experimental dataset was smaller than the reference library and reversed the spectral-matching workflow, reducing processing time from 38 to 20 minutes on the same hardware

Research Assistant

Jan 2026 – Present

UBC Department of Electrical and Computer Engineering – Lele's Lab

Hybrid

- Led the design and development of an interactive Streamlit interface for power-grid vulnerability analysis, translating researcher requirements into intuitive workflows for graph visualization, topology editing, attack comparison, and experiment tracking
- Analyzed unsuccessful graph-neural-network experiments, identified structural-camouflage failure modes, and helped redirect the project toward topology-aware robustness evaluation

Research Assistant

Sep 2024 – Present

Huazhong University of Science and Technology - Xiao Lab

Remote

- Refined a high-order bipartite graph-learning model and built training and evaluation pipelines across eight multi-view clustering benchmarks, achieving leading results on most datasets and a 6.9-point ACC gain over the strongest baseline on YouTubeFace50

- Developed DeepEP-AFDM by unfolding Expectation Propagation into trainable PriorNet and DampNet modules, reducing SER by approximately $10\times$ versus classical EP at 20 dB for 16-QAM while preserving $O(LN^2)$ complexity
- Led the research direction as corresponding author to address cross-modal misalignment and redundant feature fusion in RGB–IR detection, guiding the development of LCMA with cross-modal calibration and gated feature filtering and validating its state-of-the-art performance across three benchmarks

WORK EXPERIENCE

Technology & Innovation Assistant

UBC Faculty of Education – Learning Design & Digital Innovation

Sep 2025 – Apr 2026

Vancouver, BC

- Strengthened role-based access control for the UBC PointQuest interactive learning platform by translating instructor and student permission requirements into more secure and manageable user-access workflows
- Led the design and delivery of three AI-literacy workshops for 60+ participants, primarily educators, guiding them in the responsible use of large language models, AI-assisted presentations, and voice-synthesis tools in academic settings
- Collaborated with faculty, learning designers, and technical staff to translate pedagogical challenges into practical solutions across generative AI, multimedia production, and interactive learning technologies

Digital Information Technologies & Solutions Co-op

Sanofi

Sep 2024 – Aug 2025

Toronto, ON

- Provided cross-departmental technical support for manufacturing execution and process-information systems, coordinating with operational and technical stakeholders to maintain continuity in a pharmaceutical environment
- Performed GxP-compliant qualification and validation for more than 100 pharmaceutical technology peripherals, producing traceable documentation and helping maintain audit-ready manufacturing operations
- Built and maintained Power BI activity-monitoring dashboards that consolidated operational data and supported cybersecurity initiatives aimed at identifying unusual access patterns and reducing data-exposure risks

SKILLS

Programming	Python, Java, C++, C, SQL, R, JavaScript, HTML/CSS, React
Machine Learning	PyTorch, TensorFlow, scikit-learn, NumPy, pandas, neural networks, feature engineering, data preprocessing, generative models, VAE, diffusion models
Tools	Git, GitHub, VS Code, Jupyter Notebook, Conda, Streamlit, Power BI

AWARDS

Work Learn International Undergraduate Research Awards, UBC	2026 Summer Session
Dean’s List, UBC	2025-2026 Winter Session
Go Global International Learning Programs Award, UBC	2024 Summer Session
Dean’s List, UBC	2022-2024 Winter Session

LANGUAGES

- English: Advanced proficiency
- Chinese: Native proficiency